

Land Investment System - User Flow

Land Investment, Development & Sales Management System

User Flow

Version: 1.1 | **Companion to:** PRD v2.2 & Acceptance Criteria v1.1 | **Scope:** screen-by-screen navigation and decision points per role — not requirements, not how-to instructions

1. How to Read This Document

Each flow is shown as a sequence of screens/actions (→), with a system-triggered step shown as ↓, and a branch point marked ↘. Flows are grouped by role, then by task. For *what* each step must satisfy, see the PRD; for detailed *how-to* instructions, see the User Guide.

2. Admin — Primary End-to-End Flow

Login → Dashboard (Portfolio View) → Projects → Create Project
→ Parent Land → Landowner → Purchase Agreement → Owner Payment Schedule
→ Owner Payments → Investors → Investments → Ownership (auto-calculated)
→ Expenses → Maintenance → Plots → Cost Allocation
→ Buyer → Sale → Buyer Payment Schedule → Payments
→ Sale Completion → Profit/Loss → Investor Profit
→ Distribution Approval → Distribution → Project Closure

This is the master flow — every sub-flow below is a zoomed-in view of one segment of it.

3. Admin — Sub-Flows

3.1 Create a New Project

Dashboard → Projects → "+ New Project"
→ Enter name, location, land area → Set status (DRAFT / NEGOTIATION)
→ Save
↘ Project created, status = DRAFT

3.2 Land Acquisition

Project → Land → "+ Parent Land"
→ Enter Khata/Survey No., area, landowner details → Attach land deed
→ Save
→ Land → Purchase Agreement → "+ New Agreement"
→ Enter purchase price → Build installment schedule → Attach agreement document

- Save
- ↳ Agreement created; Balance Payable = full purchase price

3.3 Record a Landowner Payment

- Agreement → Installments tab → Select due installment
- "+ Record Payment" → Enter amount, date, method, reference → Upload receipt
- ↳ System recalculates installment status (PENDING → PARTIALLY_PAID → PAID)
- ↳ System recalculates project Balance Payable
- ↳ If method = CASH and amount ≥ statutory limit → compliance flag shown before save is allowed

3.4 Onboard an Investor

- Project → Investors → "+ Add Investor"
- Select existing investor OR create new profile (with KYC: PAN/Aadhar)
- Enter contribution amount + date → Attach investor agreement document
- ↳ Save blocked if no agreement document attached
- Save
- ↳ System recalculates Ownership % for every investor in this project

3.5 Subdivide Land into Plots

- Project → Plots → "+ Subdivide"
- Add plots (plot no., area, dimensions, facing) – one by one or bulk import
- Choose cost allocation method: Area-based / Percentage-based / Manual
- ↳ System computes each plot's allocated cost
- Review allocated costs → Publish plots as AVAILABLE

3.6 Sell a Plot or Land

- Project → Sales → "+ New Sale"
- Choose sale type: Whole Land / Plot-wise / Multiple Plots / Mixed
- Select or create buyer (with KYC)
- Add Sale Item(s) → plot(s)/parcel + agreed price
- Enter circle-rate/DLC reference value
- ↳ If sale price < circle rate → compliance flag shown before save
- Build buyer payment schedule (down payment + installments)
- (Optional) Add agent + commission rule
- Save → Sale status = SALE_AGREEMENT, Plot status = BOOKED
 - ↳ Registration No. + stamp-duty status recorded → Sale status → SOLD
 - ↳ Not yet recorded → Sale stays at SALE_AGREEMENT

3.7 Record a Buyer Payment

- Sale → Installments tab → Select due installment
- "+ Record Payment" → Enter amount, date, method, reference → Upload receipt
- ↳ System recalculates installment status and Balance Receivable
- ↳ Installment overdue → auto-flag OVERDUE → Buyer notified, Admin Overdue widget updated

3.8 Calculate Profit & Loss

- Project → Profit & Loss → "Calculate"
- ↳ System derives Net Sale Proceeds, Actual Project Cost, Project Profit/Loss

- ↳ Marked "Forecast" → visible for planning only, cannot feed a Distribution
- ↳ Marked "Final" → becomes the base for Distribution

3.9 Record & Pay a Distribution

- Profit & Loss (Final) → Distributions → "+ New Distribution"
- ↳ System computes each investor's payout (Ownership % × Distributable Profit)
- Admin saves it directly – no approval/review step, immediately available for payment
- Mark each investor payout as paid (+ proof)
- ↳ System generates investor statement automatically; each investor is notified (WhatsApp, + SMS if enabled)
- ↳ Proposed amount > distributable profit → save is rejected (data-integrity check, not an approval)

3.10 Close a Project

- All plots SOLD + all payments reconciled + all distributions settled
- Project → "Close Project"
- ↳ Project status → CLOSED
- ↳ Project becomes read-only for financial edits; remains visible in reports/history

4. Member Flow

The system has exactly two login roles (PRD §5). Member is a single, simpler role — there is no separate Project Manager or Accountant login anymore.

- Login → Dashboard
- Projects → View list (read-only – no "+ New Project", no edit, no delete)
- Sales (Income) → "+ New Sale Entry" → record a buyer payment (same pattern as §3.7) → Save
- Purchases (Expenses) → "+ New Purchase Entry" → record a landowner payment or expense (same pattern as §3.3 / §3.6) → Save
- Sales / Purchases → View lists, search, filter
- ↳ Investors → not shown in navigation at all for Member – no route, no data, no partial view
- ↳ Project create/edit/delete, Plot management, Distribution, Settings → not available to Member; shown as Admin-only where visible at all, hidden entirely where they'd otherwise leak investor-adjacent data

5. Cross-Role Flow — Sale Payment to Distribution

This flow shows how one event (a buyer payment, entered by either role) ripples through the system without manual re-entry, and without any approval step:

- Sale payment recorded (by Admin or Member)
 - ↳ Payment Transaction created (immutable)
 - ↳ Installment status updates; Balance Receivable recalculates
 - ↳ Financial Ledger entry created
 - ↳ WhatsApp notification sent to the buyer (+ SMS if enabled); Admin/Member dashboard reflects updated balance
 - ↳ (Later) Admin runs Profit & Loss "Final" calculation → this payment's sale now counts toward Realized Profit
 - ↳ Admin records a Distribution directly (§3.9 – no approval step) → Investor Profit computed from Distributable Profit
 - ↳ Each investor is notified (WhatsApp + SMS if enabled) and can be sent their statement –

investors have no login of their own, so this notification/statement is their only view into it

6. Notification Flow

Every relevant action notifies whichever party is concerned — a system user (Admin/Member) or a non-login party (Buyer, Landowner, Investor) reached via their stored phone number. This is the same underlying trigger mechanism regardless of who or what triggers it:

Trigger event (payment recorded, installment overdue, distribution recorded, project status change, etc.)

- ↓ System resolves the concerned party's contact details
- ↓ WhatsApp message sent (always – mandatory channel)
- ↳ SMS also enabled in this deployment's configuration → SMS sent too
- ↳ SMS not enabled/purchased → skipped entirely, silently, no error, no charge
- ↓ Delivery outcome logged
- ↳ Send fails (either channel) → logged for retry; the underlying business record (payment/sale/distribution) is never rolled back or blocked because of a notification failure

7. Edge-Case Flows

7.1 Overdue Installment (Buyer or Landowner side)

Due date passes with no payment recorded

- ↓ Daily status job runs → Installment status → OVERDUE
- ↓ Concerned party notified via WhatsApp (+ SMS if enabled); Admin/Member Overdue widget updated
- ↳ Admin/Member sends reminder OR records a late payment directly (\$3.3 / \$3.7)

7.2 Sale Cancellation (Admin only)

Sale (any status before SOLD) → "Cancel Sale"

- Confirm cancellation
- ↓ Plot(s) revert to AVAILABLE
- ↓ Sale status → CANCELLED
- ↓ Any prior payment transactions → VOIDED (not deleted), linked audit entry created – direct Admin action, no approval step
- ↳ Plot becomes sellable again

7.3 Refund (Admin only)

Original Payment Transaction → "+ Refund"

- Enter refund amount, date, reason
- ↓ System creates a linked REFUND transaction (original transaction remains in history, unedited)
- ↓ Receivable/Payable and Ledger recalculate

7.4 Investor Exit Mid-Project (process to be confirmed — see PRD §18, Open Question 1)

Investor requests exit (communicated to Admin outside the system, since Investor has no login)
 → Admin records the outcome directly (no approval step)
 ↳ Share transfer to another investor → new ProjectInvestor record + audit entry + updated Ownership % for all investors
 ↳ Buyout by the project itself → treated as a Settlement (\$7.5 of PRD) with an early-exit adjustment

8. Flow-to-Screen Quick Reference

Flow	Primary Screens Touched	Who
Create Project	Projects List → New Project Form	Admin only
Land Acquisition	Land → Parent Land Form → Purchase Agreement Form	Admin only
Landowner Payment (Purchases)	Purchase Agreement / Purchases → Installments → Payment Entry	Admin, Member
Investor Onboarding	Investors → Add Investor Form → Investment Entry	Admin only — screen not shown to Member
Plot Subdivision	Plots → Subdivide Form → Cost Allocation Review	Admin only
Sale Creation	Sales → New Sale Form → Sale Items → Payment Schedule Builder	Admin only (Member can record payments against an existing sale, not create one — confirm against PRD if Member should also create sales)
Buyer Payment (Sales)	Sale / Sales → Installments → Payment Entry	Admin, Member
Profit Calculation	Profit & Loss → Calculate	Admin only
Distribution	Distributions → Record → Payout Entry (no approval step)	Admin only
Project Closure	Project → Close Project	Admin only
Investor Statement	Admin generates/sends from Investor Detail — no investor login/dashboard exists	Admin only
Buyer Payment History	Sales → Purchase Detail → Payment History	Admin, Member

The Sale Creation row flags a genuine gap in what you specified: “Member can create entries” is stated broadly, but the worked examples you gave were Sales/Purchases *payments*, not full Sale Agreement creation (which involves picking plots, pricing, and a payment schedule — closer to project-setup than a simple entry). I’ve defaulted to Sale Agreement creation being Admin-only and Member limited to recording payments against it, consistent with Member’s read-only stance on Plots and Projects — confirm if Member should be able to originate a new sale outright.